

Kubernetes environment setup

Step 1: Install kubectl

PowerShell (Run as Administrator):

Command: `winget install Kubernetes.kubectl`

Verify:

Command: `kubectl version --client --output=yaml`

Step 2: Install Docker Desktop

[Docker Desktop](#) – download & install

<https://docs.docker.com/desktop/setup/install/windows-install/>

Verify:

Command: `docker version`

Step 3: Install Minikube

PowerShell:

Command: `winget install Kubernetes.minikube`

Verify:

Command: `minikube version`

Step 4: Start Minikube

Command: `minikube start --driver=docker`

Step 5: Verify Cluster

Command: `kubectl get nodes`

Expected:

NAME	STATUS	ROLES	AGE
minikube	Ready	control-plane	

Minikube + Docker Architecture (Windows)

